
Microglial Classification Using Deep Learning Methods

Isha Rai

Department of Biomedical Engineering
New Jersey Institute of Technology
Newark, NJ 07102
ir76@njit.edu

Vishesh Raju

Department of Computer Science
New Jersey Institute of Technology
Newark, NJ 07102
vr432@njit.edu

Abstract

Microglial cells undergo morphological changes in response to neurological injury, making their classification essential for studying brain pathology. However, manual labeling of microglial morphologies is costly, subjective, and limited by severe class imbalance in biological datasets. This project presents a deep learning framework for automated microglial morphology classification using supervised and unsupervised approaches.

1 Introduction

Deep learning network architectures can be applied to a wide variety of real-world applications, with the intent of reducing bias and increasing throughput. For this project, a solution to many neuroscience study limitations has been proposed. Traumatic brain injury research is reliant on animal models to understand the robust characteristics of the brain. Specifically, understanding the cellular responses of the brain's main building blocks: neurons, microglia, astrocytes, and oligodendrocytes is of utmost priority.

Microglial cells have been shown to take on morphological (structural) changes in response to infection and injury which some researchers believe may influence the clinical outcomes of patients. In the case of traumatic brain injury, microglia in animal models are photographed using biochemical methods of fluorescence. Once cells are segmented from each image, analysis techniques can capture morphology changes, if any.

Once such field of traumatic brain injury are blast injuries, primarily found in soldiers and other's exposed to high level blast waves. Microglial response to blast injuries is more nebulous. Since there is no focal point of injury, the damage is widespread and multi-phasic so some microglial cells may appear to be healthy while its neighbor may appear to be under attack. Our proposed network architecture aims to classify imaged microglia into one of five published shape modes, each representing a different activation state of the cell.

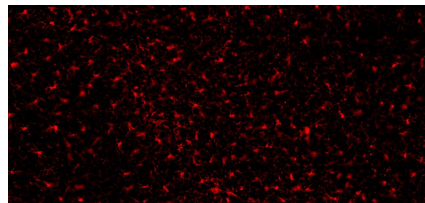


Figure 1: Sample microglial image (before processing)

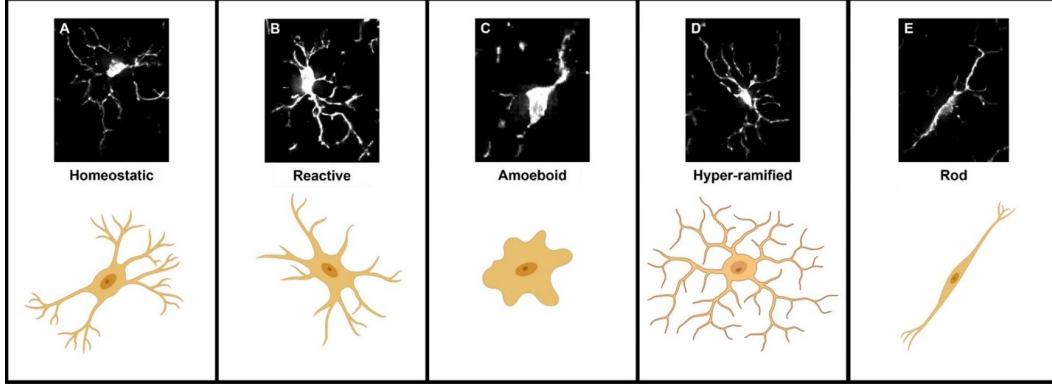


Figure 2: Representations of the five shape modes observed with sample cells[1]

1.1 Contributions

The following outlines the contributions made by each team member to this project.

- **Isha Rai:** Dataset acquisition, Data Cleaning, Data Analysis, Cell Image Segmentation, Cell image labelling.
- **Vishesh Raju:** Supervised learning, Unsupervised learning, clustering analysis, data augmentation techniques.
- **Both authors:** Model evaluation, interpretation of results, and report writing.

2 Dataset Description

The dataset for this project was sourced from images obtained at the CIBM3 lab at NJIT. Brain tissue samples were stained with IBA1, a microglial marker, and the resulting fluorescence microscopy images were processed using ImageJ for segmentation. From these images, a total of 323 microglial cells were extracted. For the supervised learning pipeline, the cells were manually labeled by an expert. The resulting dataset is heavily biased toward certain morphologies, particularly the reactive and amoeboid shape, reflecting the inherent class imbalance present in biological samples. Later, the dataset was balanced using data augmentation techniques such as rotation, flipping, and adjustments to brightness and contrast to a total of 630 cells(126 each shape mode).

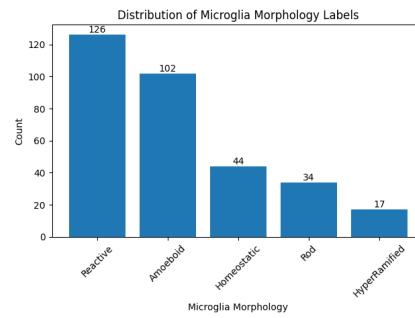


Figure 3: Distribution of labelled cells

The dataset for this project comprises three distinct modalities for each microglial cell. First, the raw cell images obtained from fluorescence microscopy provide the visual data of individual microglia. Second, the skeletonized images, processed using ImageJ, highlight the cellular morphology and branching structure, which are critical for feature extraction and shape analysis. Third, the skeleton feature CSV files contain quantitative measurements extracted from the skeletonized images, including branch lengths, number of endpoints, and other morphometric features. Together, these three modes of data raw images, skeletonized images, and skeleton features enable both supervised and unsupervised deep learning pipelines, supporting classification, clustering, and morphological analysis of microglial cells.

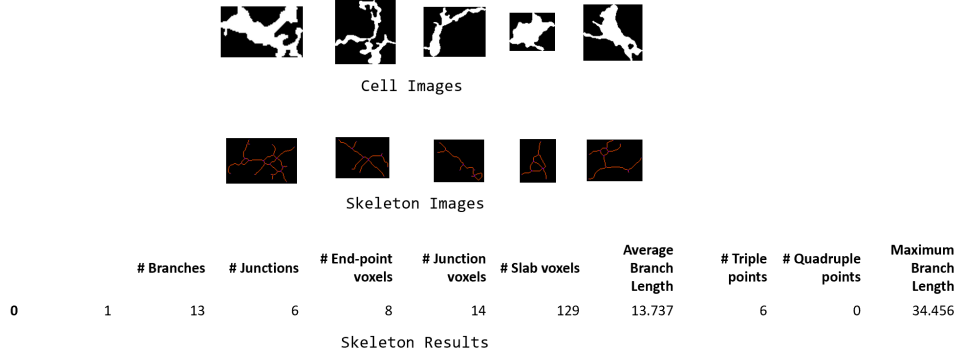


Figure 4: Data Modalities

The microglial cell images were categorized into five morphological groups based on their activation state: Homeostatic, Reactive, Amoeboid, Hyper-ramified, and Rod, as illustrated in Figure 2. These labels were assigned by expert annotation and are used for the supervised learning pipeline. Each cell image corresponds to one of these groups, providing ground-truth information for training and evaluating classification models.

3 Methodology

This section presents the overall methodological framework, covering data preprocessing, model architectures, training strategies, and evaluation procedures for microglial morphology classification.

3.1 Supervised Learning

For the supervised learning pipeline, we employed a multimodal convolutional neural network (CNN) that leveraged all three data modalities: raw cell images, skeletonized images, and skeleton feature CSV files of original unbalanced dataset with 323 cells. The network was trained using expert-labeled data, where each cell was assigned to one of the five morphological groups. By integrating information from the three modalities, the multimodal CNN aimed to capture both visual and morphometric features of microglia, enabling accurate classification across all five shape categories.

3.2 Unsupervised Learning

3.2.1 Standard Unsupervised Run

The initial unsupervised experiments used a multimodal autoencoder trained on the original 323-cell dataset without any class balancing. The autoencoder compressed the raw images, skeletonized images, and skeleton feature CSVs into a 64-dimensional latent space, which was then clustered using K-means with $k = 5$. The resulting clusters showed a strong bias toward dominant morphologies, with cluster 2 containing 49.2% of the cells while other clusters were underrepresented. This run highlighted the limitations of the original dataset for unsupervised analysis and the need for balancing strategies.

3.2.2 Unsupervised Run with Balanced Dataset

To reduce class imbalance, additional cell images were incorporated to create a balanced dataset of 630 samples. Using the same multimodal autoencoder and K-means clustering pipeline, the cluster distribution became more uniform across the five clusters, with percentages ranging from 9.7% to 30.3%. This demonstrates that balancing the dataset through augmentation improves cluster separability and allows the unsupervised model to better capture underrepresented morphological types.

3.2.3 Denoising Autoencoder Approach

In the final iteration, a denoising autoencoder was applied to the original dataset to enhance latent feature extraction. Noise was injected into the input images, which were then reconstructed through the autoencoder before clustering with K-means. This approach improved cluster definition and representation, with cluster 4 accounting for 41.5% of cells while the remaining clusters were more evenly distributed than in the original run. The denoising strategy highlights the potential for noise-resistant representations that capture essential morphological features even in small, biased datasets.

4 Evaluation Metrics

4.1 Supervised Learning

This set was heavily biased toward the amoeboid morphology, so the results were unsatisfactory. The confusion matrix is illustrated below in Figure 5, showing very high confidence on amoeboid-like cells, but little to no confidence on other cell types.

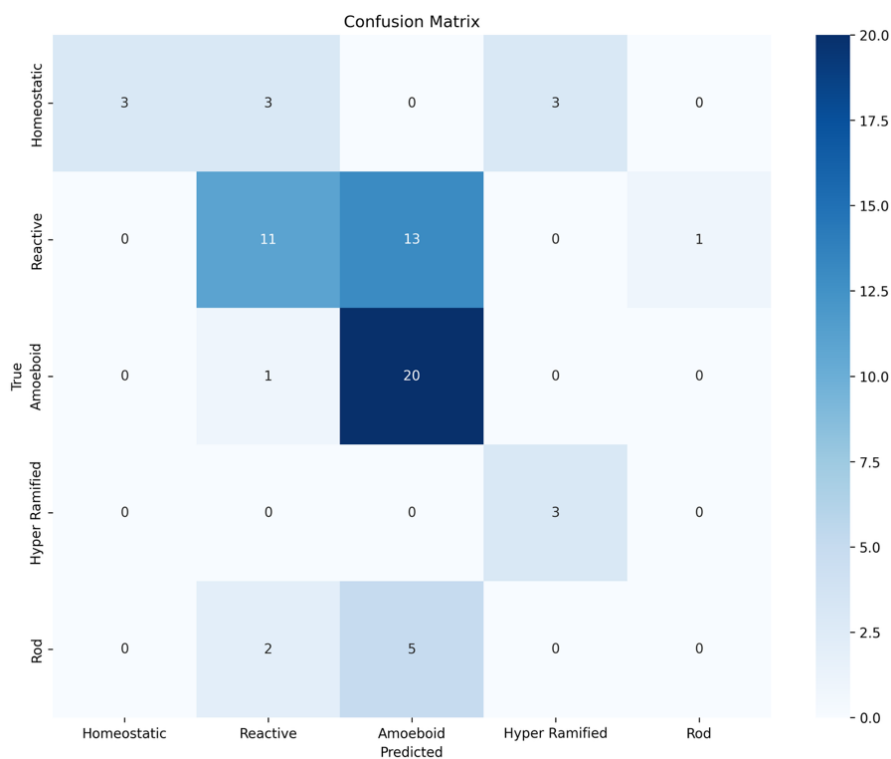


Figure 5: Confusion Matrix

Training and validation loss and accuracy illustrated in Figure 6 represent this bias, with spikes in validation loss and accuracy around epoch 25. This spike shows unstable generalization, which is contrary to the main goal of our project. Training accuracy stagnating early in the epochs also shows strong memorization by the model

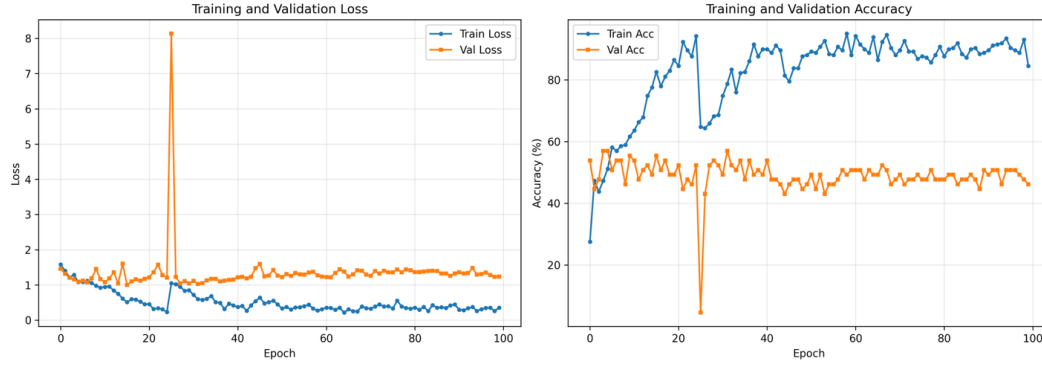


Figure 6: Training and Validation Loss & Accuracy

4.2 Unsupervised Learning

Our next iteration aimed to use unsupervised learning, hoping to minimize some effects of the biased dataset. For this iteration, we also used techniques to determine if five shape modes were ideal in the input data. In application, understanding the ideal number of shape modes will help us identify if all cell types are equally represented in training phases, or if more data should be added before running the framework.

4.2.1 Standard Unsupervised Run

Cluster visualization below shows high-dimensional latent representations of each data point to two principal components. Cluster two forms a dense, highly populated region with few outliers, revealing high levels of similarity in these cells. Clusters 0, 1, 3, and 4 occupy more peripheral locations, also a good sign, showing these cells contain unique latent features.

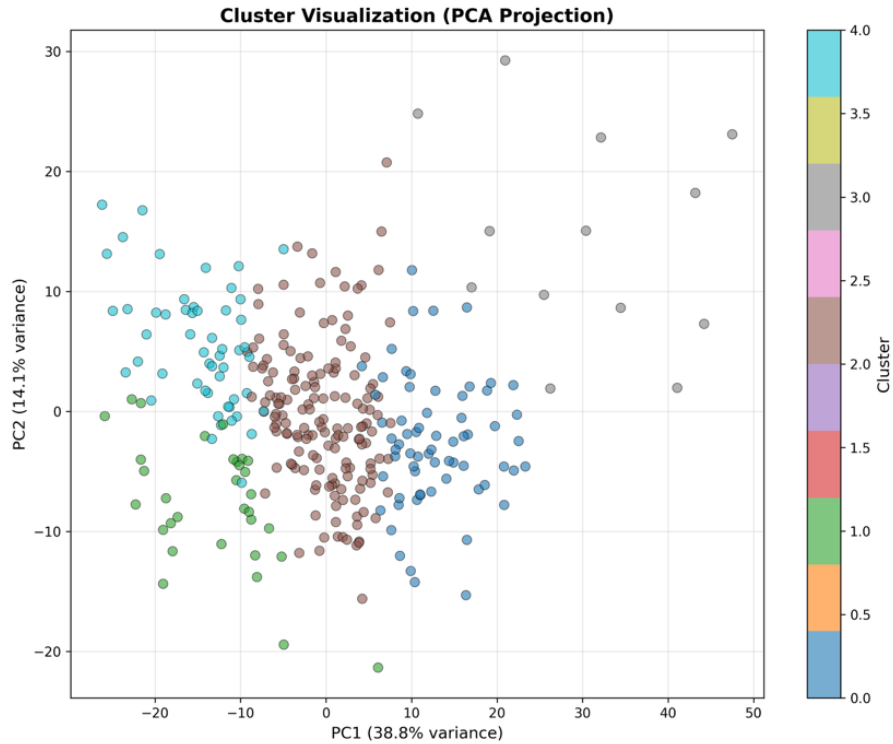


Figure 7: PCA Projection

In this unsupervised model, the dataset bias is further confirmed. The cluster distribution bar chart shows how many of each cell type was found by the model in the data. A positive sign is the similarity between the clusters in this model compared to the clusters in the supervised model, which validates unsupervised learning against expert labeling.

Below is shown an analysis of the number of clusters chosen. Though five clusters carry biological importance, the elbow curve reveals only three clusters to be the optimal value. After three, the sum of squares within each cluster does not vary significantly. The same value is confirmed by the silhouette score, where the highest score is seen at three clusters.

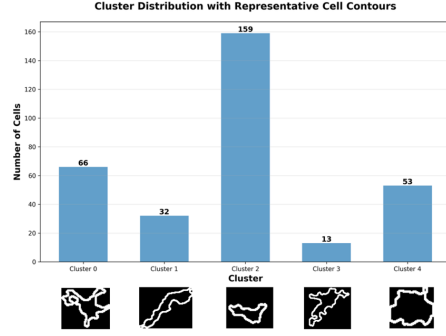


Figure 8: Cluster Distribution

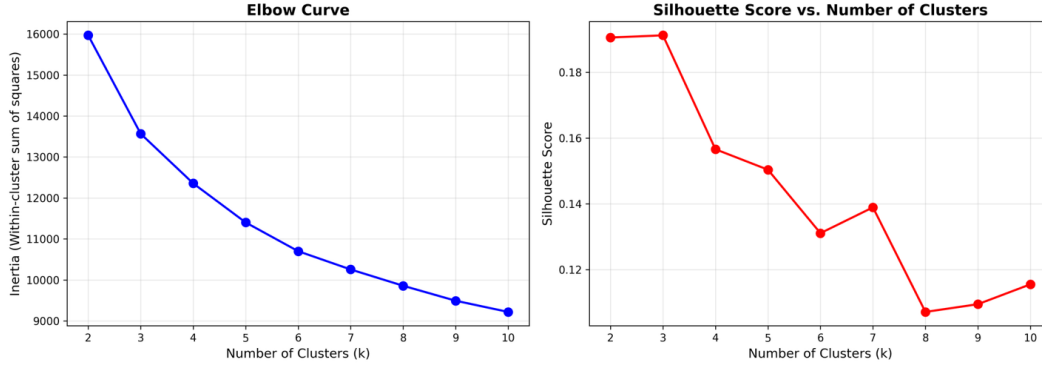


Figure 9: Elbow Curve and Silhouette Score

4.2.2 Unsupervised Run with Balanced Dataset

At this phase, we began to consider utilizing data augmentation techniques covered in the curriculum to balance the number of cells in less represented clusters in the original dataset. We implemented rotations, flips, brightness, contrast, and noising methods to certain cells in hopes of balancing the dataset. Each image was modified using only one augmentation method. The labels were carried over to augmented images. we tested the model once again with a total of 620 cells in which the dataset was less biased toward any one shape mode. Some bias was still present, due to the representation of certain shape modes in the biological data.

The cluster visualization here tended to be less ideal than the previous iteration. Though some of the clusters are more precise in comparison to other data points in their respective clusters, many datapoints lie outside of groups. Overlaps in clusters also show confusion in sorting. Importantly, the true labels (expert labels) also showed nearly the exact same pattern, which may indicate presence of latent features that cannot be separated by our model or the human eye.

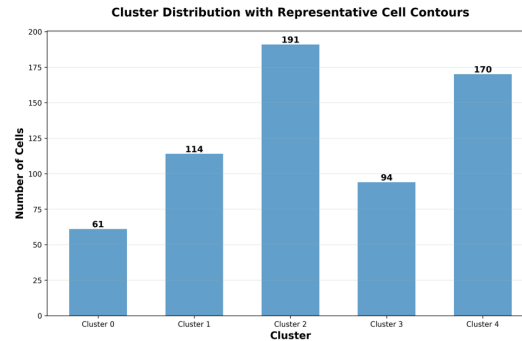


Figure 10: Cluster Distribution (Balanced)

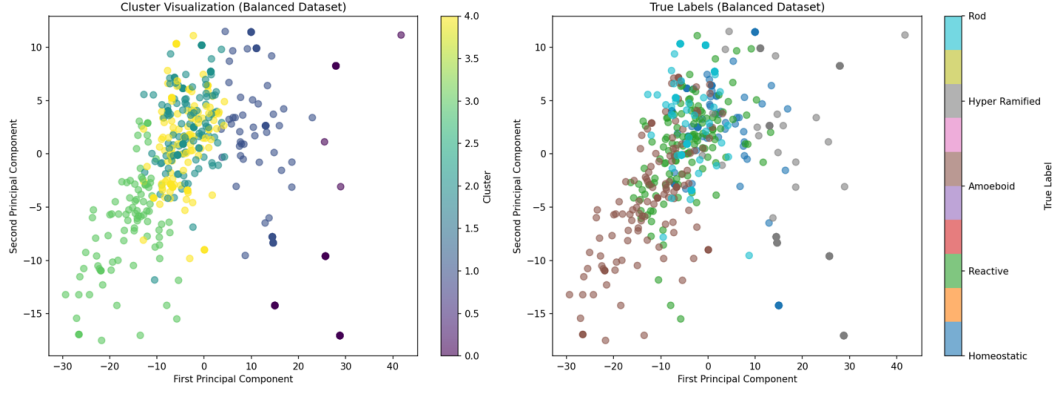


Figure 11: PCA Projection (Balanced Dataset)

We present again an elbow curve, which in this iteration suggests more iterations can be differentiated when compared to this model on the first dataset. The silhouette score on the contrary, suggests that only two clusters contain high inertia.

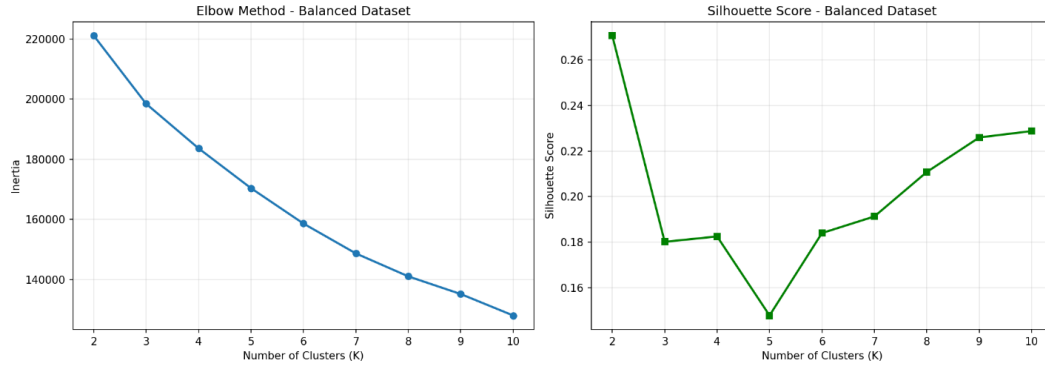


Figure 12: Elbow Curve and Silhouette Score (Balanced Dataset)

Following these steps, the results were poor, and upon closely analyzing skeletonized images of augmented cells, we realized that many of the features used for clustering became nonsensical after augmentation.

4.2.3 Denoising Autoencoder Approach

The cluster distribution obtained from the denoising autoencoder shows a more structured allocation of samples across the five clusters compared to the standard unsupervised run. Cluster 4 contains the largest number of cells (134 samples), while the remaining clusters range from 11 to 67 samples. Although some imbalance persists, the denoising strategy reduces extreme dominance by a single cluster and improves representation of previously underrepresented morphologies. This distribution suggests that the denoising autoencoder learns more meaningful latent features that better capture structural diversity among microglial cells. The PCA-based visualization of the latent space learned by the denoising autoencoder reveals clearer cluster

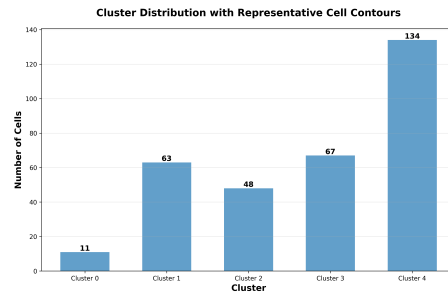


Figure 13: Cluster Distribution (Denoising)

separation compared to earlier unsupervised experiments. While overlap remains due to biological variability, several clusters form compact groupings, indicating improved feature organization. When compared with the true expert labels, partial alignment between unsupervised clusters and known morphological categories is observed. This highlights that the denoising autoencoder captures biologically relevant features despite not being trained on labels, supporting its effectiveness for exploratory morphological analysis.

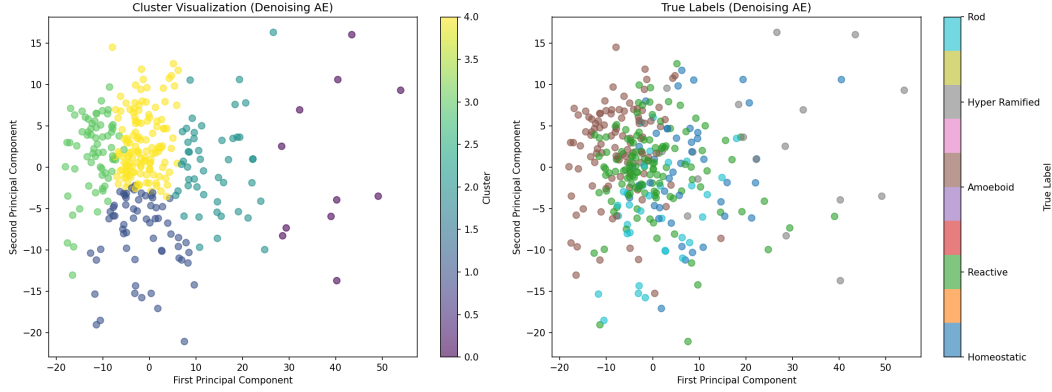


Figure 14: Cluster Distribution (Denoising)

The elbow method applied to K-means clustering on the denoised latent space shows a clear change in slope around $k = 5$, supporting the selection of five clusters as a balance between compactness and interpretability. This choice aligns with the expert-defined morphological categories. Silhouette score analysis further supports this selection, with reasonable scores maintained at $k = 5$, indicating moderate cluster separation. Although some overlap remains due to biological variability, the denoising autoencoder demonstrates improved cohesion and robustness compared to the standard unsupervised baseline, making it well-suited for clustering in small, noisy biological datasets.

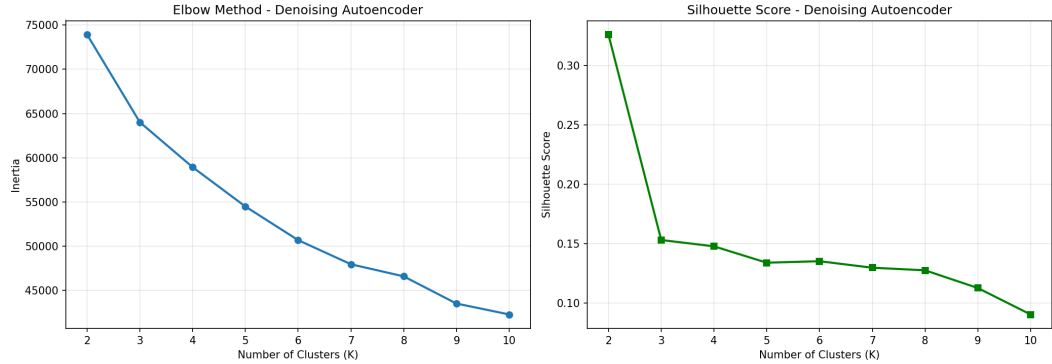


Figure 15: Elbow Curve and Silhouette Score (Denoising)

5 Conclusion

This project investigated both supervised and unsupervised deep learning approaches for microglial morphology analysis using multimodal data derived from fluorescence microscopy images. The supervised multimodal CNN achieved a best validation accuracy of 56.92%; however, its performance was limited by the small dataset size and severe class imbalance, leading to overfitting and reduced generalization. In contrast, the unsupervised learning framework, particularly the denoising autoencoder combined with K-means clustering, demonstrated strong capability in discovering meaningful morphological groupings without relying on expert labels.

Table 1: Comparison of Supervised and Unsupervised Experimental Runs

Run	Approach	Dataset	Samples	Key Results
Unsupervised	Autoencoder + KMeans	Original	323	49.2% Cluster purity
Unsupervised (Balanced)	Autoencoder + KMeans	Augmented	630	41.6% Cluster purity
Denoising AE	Denoising AE + KMeans	Original	323	51.8% Cluster Purity
Supervised	Multimodal CNN	Original	323	Best Val Acc: 56.92%

Among all experimental runs, the unsupervised approaches proved to be more robust and unbiased, as they were not influenced by labeling subjectivity or dataset imbalance. The balanced and denoising variants further improved cluster quality and latent feature representation, indicating that unsupervised learning is well-suited for exploratory analysis of complex biological data. Overall, this study shows that unsupervised, bias-free deep learning methods provide a more reliable and scalable solution for microglial morphology analysis, especially in scenarios where labeled data is limited or inherently imbalanced.

References

- [1] Reddaway, J., et al. Microglial morphometric analysis: so many options, so little consistency. *Frontiers in Neuroinformatics*, vol. 17, 2023.